

EECE7352 Computer Architecture

Homework 3

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Part A: (10 Points)

For any two current embedded processors (e.g., ARM7, ARM9, AMD Ryzen Embedded, TI MSP430, MIPS32, Apple A15, RISC-V E7 or S7), provide the details of the microarchitecture of the pipeline (e.g., the width of the data path, the number of pipeline stages, number of instructions issued per cycle, the number of integer units, the branch resolution unit, the names of the pipeline stages, etc.). You might not find every one of these details but provide at least 3 different pipeline characteristics that you can compare between two embedded CPUs. Make sure to cite your sources.

Ans:

For the comparison of two current Embedded Processors, I have taken **ARM Cortex M7** and **SiFive E76 RISC-V** Core Processors. Both are 32-bit, dual issue, in-order RISC microcontrollers aimed at High Performance Embedded and Automotive Applications

1. ARM Cortex-M7 Architecture (ARM DDIO489B)

- **Width of the Datapath:** 32-bit integer core and an optional 64-bit FPU
- **ISA:** ARMv7-M (Proprietary ARM RISC)
- **Pipeline Stages:** Total of '6' Stages in the pipeline.
 - **IF1** -> Instruction Fetch 1
 - **IF2** -> Instruction Fetch 2 (this is the prefetch buffer)
 - **ID** -> Instruction Decode
 - **EX1** -> Execute 1
 - **EX2/MEM** -> Execute 2 / Memory Access
 - **WB** -> Write-back
- **Instruction per cycle:** Dual issue in-order pipeline (2 instructions per cycle)
- **Integer Units:** Each processor has 2 ALU's (one ALU for each pipe) and additionally has 1 Dedicated Multiply Accumulate (MAC), 1 Divider, 1 Load Store Unit (LSU).
 - **Total:** 2 ALU + 1 MAC + 1 Divider + 1 LSU
- **Branch Penalty:** 1-2 Cycles
- **Clock Freq:** < 400 MHz

2. RISC-V E7 Series (SiFive E76)

- **Width of the Datapath:** 32 -bit integer core
- **ISA:** RISC-V RV32IMAFIC (Open RISC-V RISC core)
- **Pipeline Stages:** Total of '8' Stages in the pipeline.
 - **F1** -> Instruction Fetch 1
 - **F2** -> Instruction Fetch 2
 - **D1** -> Decode 1
 - **D2** -> Decode 2
 - **AG** -> Address Generation
 - **M1** -> Memory Access 1
 - **M2** -> Memory Access 2
 - **WB** -> Write-back
- **Instruction per cycle:** Dual issue in-order pipeline (2 instructions per cycle)
- **Integer Units:** Each processor has 2 integer pipelines, 1 fully pipelined integer multiplier, 1 integer divider and 1 Load store Unit
 - **Total:** 2 integer pipelines (A & B) + 1 Multiplier + 1 Divider + 1 LSU
- **Branch Penalty:** 4-6 Cycles
- **Clock Freq:** 500 – 600 MHz

Key Insights:

- Both the processors are 32-bit, dual issue, in-order RISC pipelined with Harvard system Architecture (separate memory spaces for data and address)
- Since the pipeline depth is lower in Cortex-M7 than E76, it is better suitable for hard-real time applications, whereas E76 is used for processes needing a better throughput.
- E76 having a deeper pipeline supports higher frequencies compared to Cortex M7

Citation:

1. **ARM. (2014). ARM Cortex-M7 Processor Technical Reference Manual (Rev. r0p2, ARM DDI 0489B).**
2. **SiFive, Inc. (2020). SiFive E76 Manual (Ver. v19.08p3p0).**